



GulfZone HR Management System

Deployment Guide for Windows Server

Next.js application + cloud-hosted Supabase • IIS reverse proxy • PM2 process manager

This guide walks an IT administrator through publishing the GulfZone HR application on an on-premises Windows Server. The database, authentication, and file storage remain in cloud Supabase; the Windows Server runs the web application and serves it securely over HTTPS.

Architecture

Internet → **IIS** (HTTPS 443, SSL, domain, reverse proxy)

- forwards **/HRportal** → **Node.js / Next.js** on localhost:3000 (managed by PM2)

- outbound HTTPS → **Supabase** (cloud database, auth, storage)

Prerequisites

Install these on the Windows Server once before you begin.

Component	Where to get it	Purpose
Node.js LTS v20+	nodejs.org (Windows MSI)	Runs the application; includes npm
IIS	Windows feature (Web Server role)	Front-end web server / SSL
URL Rewrite 2.1	iis.net/downloads	IIS module for routing
Application Request Routing (ARR)	iis.net/downloads	Enables IIS reverse proxy
SSL certificate	Your CA or Let's Encrypt (win-acme)	HTTPS for the domain

■ **Note:** After installing ARR, enable proxying: **IIS Manager** → **server node** → **Application Request Routing Cache** → **Server Proxy Settings** → tick **“Enable proxy”** → **Apply**.

1 Copy the application to the server

- Copy the project folder to a stable location, for example **C:\apps\GulfZoneHR**.
- Confirm package.json and next.config.js are present in that folder.

2 Configure environment variables

Copy the template and fill in your real Supabase values:

```
copy deploy\windows\.env.local.example .env.local
notepad .env.local
```

Set these keys (from Supabase Dashboard → Project Settings → API):

Variable	Value
NEXT_PUBLIC_SUPABASE_URL	https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY	anon / publishable key
SUPABASE_SERVICE_ROLE_KEY	service role key (server only)
NEXT_PUBLIC_BASE_PATH	/HRportal
NEXT_PUBLIC_APP_URL	https://hr.clientdomain.com

■ **Note:** Values starting with **NEXT_PUBLIC_** are compiled into the build. Always set them **before** building, and rebuild if you change them later.

3 Install PM2 (keeps the app running + auto-start on reboot)

```
npm install -g pm2 pm2-windows-startup
pm2-startup install
```

4 Build and start the application

From the project root, run the bundled deploy script:

```
cd C:\apps\GulfZoneHR
deploy\windows\deploy.bat
```

This installs dependencies, builds the production bundle, and starts the app under PM2. Verify it is running:

```
pm2 status
```

The app now listens locally on `http://localhost:3000/HRportal`.

5 Configure IIS reverse proxy (HTTPS + domain)

- Create or select an IIS site bound to your domain on **port 443** with the SSL certificate.
- Copy `deploy\windows\web.config` to the site root (e.g. `C:\inetpub\wwwroot\web.config`).
- In **IIS Manager** → **URL Rewrite** → **View Server Variables**, allow `HTTP_X_FORWARDED_HOST` and `HTTP_X_FORWARDED_PROTO`.
- Restart the site.

✓ The application is now live at: <https://hr.clientdomain.com/HRportal>

6 Firewall

- **Inbound:** allow TCP **443** (and **80** for the HTTP→HTTPS redirect).
- **Outbound:** allow TCP **443** so the server can reach Supabase.
- Do **not** expose port 3000 publicly — only IIS talks to it locally.

7 Create the login accounts (first deployment only)

If you provisioned a fresh Supabase project from the deployment SQL, the database rows exist but Auth passwords do not. Create the login accounts one of two ways:

- Use the app's **Admin** → **Employees** → **Add Employee** flow (creates the auth user + temporary password), or
- Supabase Dashboard → **Authentication** → **Add user** (same email).
- Then enable **Authentication** → **Providers** → **Leaked password protection**.

Day-2 Operations

Task	Command
Check status	<code>pm2 status</code>
View live logs	<code>pm2 logs gulfzone-hr</code>
Restart the app	<code>pm2 restart gulfzone-hr</code>
Stop the app	<code>pm2 stop gulfzone-hr</code>
Deploy an update	copy new files, then <code>deploy\windows\deploy.bat</code>
After a server reboot	Automatic — PM2 relaunches the app

Updating to a New Version

- Copy the new application files over the existing folder (keep `.env.local`).
- Run `deploy\windows\deploy.bat` — it rebuilds and restarts automatically.

Troubleshooting

Symptom	Likely cause	Fix
502 / 504 from IIS	Node not running, or ARR proxy off	Run <code>pm2 status</code> ; enable ARR proxy
Blank page / assets 404	basePath mismatch	URL must include <code>/HRportal</code> ; rebuild after env change
Login fails / session drops	Site not served over HTTPS	Bind SSL; force HTTPS (web.config rule)
“supabaseUrl is required”	Missing / un-built env vars	Check <code>.env.local</code> , then rebuild
Works on <code>:3000</code> but not domain	IIS rewrite / ARR not set	Re-check Step 5

Files Included in `deploy\windows`

File	Purpose
<code>web.config</code>	IIS reverse-proxy + HTTPS + security headers
<code>ecosystem.config.js</code>	PM2 process definition (auto-restart, logs)
<code>deploy.bat</code>	One-click build + (re)start script for updates
<code>.env.local.example</code>	Template for production environment variables
<code>WINDOWS_DEPLOYMENT.md</code>	Text version of this runbook

Note: This deployment keeps the database, authentication, and document storage in cloud Supabase. The Windows Server requires reliable outbound internet (HTTPS) to function. For a fully on-premises database, Supabase can be self-hosted via Docker — a larger undertaking outside this guide.